

Research Validation & Industry Evidence

Why This Problem Matters

The recruitment industry was built on the assumption that resumes are reliable indicators of candidate quality.

However, the rise of generative AI has fundamentally changed this assumption.

Today, candidates can use ChatGPT and other AI tools to:

- Optimize resumes for ATS systems
- Match job description keywords
- Rewrite professional experience
- Generate compelling cover letters

As a result, employers face an increasing challenge in distinguishing between:

- Candidates who can write a good resume
- Candidates who can actually perform the job

This creates a growing need for capability-based assessment rather than document-based screening.

Research Evidence Supporting Our Approach

Research 1: Work Sample Assessments Are Strong Predictors of Job Performance

One of the most influential findings in industrial-organizational psychology is that work sample assessments are among the strongest predictors of actual job performance.

According to Schmidt & Hunter's landmark meta-analysis covering 85 years of personnel selection research, work sample tests achieved a predictive validity of approximately 0.54, outperforming many traditional hiring methods including interviews and personality tests. This means candidates who perform well in realistic job tasks are significantly more likely to perform well on the job.

Reference: <https://www.cogn-iq.org/blog/work-sample-tests/>

Supporting Research: <https://home.ubalt.edu/tmitch/645/articles/A%20META-ANALYSIS%20OF%20WORK%20SAMPLE%20TEST.pdf>

Key Insight:

"The best predictor of future job behavior is observed behavior in a similar situation."

This directly supports our capability-first hiring model.

Research 2: Game-Based Assessments Can Measure Real Cognitive Ability

A large-scale validation study involving more than 11,000 assessment completions demonstrated that game-based cognitive assessments can reliably measure cognitive abilities while maintaining fairness and improving candidate experience.

The study found:

- Convergent validity of approximately 0.50
- Test-retest reliability of approximately 0.68
- Positive candidate experience
- Reduced test anxiety

Reference: <https://www.frontiersin.org/articles/10.3389/fpsyg.2022.942662/full>

Key Insight:

Game-based assessments can successfully evaluate logical thinking and cognitive ability while providing a more engaging candidate experience.

Research 3: Systematic Reviews Support Game-Based Hiring Assessments

A systematic review published in Frontiers in Psychology reviewed game-related assessments used for personnel selection.

Researchers concluded that game-based assessments have strong potential for:

- Measuring cognitive ability
- Evaluating behavioral traits
- Improving applicant experience
- Reducing candidate disengagement

Reference: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9554090/>

Key Insight:

Assessment games are most effective when they evaluate job-relevant behaviors rather than entertainment-focused gameplay.

Research 4: Behavioral Assessments Reveal Different Information Than Self-Reported Answers

A 2026 study comparing game-based problem-solving assessments with self-report questionnaires found that behavioral performance and self-reported confidence often do not align.

Reference: <https://arxiv.org/abs/2602.05525>

Key Insight:

People frequently overestimate or underestimate their own capabilities.

Observing actual behavior provides a different and potentially more valuable signal than asking candidates to describe themselves.

Existing Market Validation

Several global companies have already proven market demand for game-based and simulation-based hiring assessments.

HireVue

Uses AI-driven assessments and game-based evaluations to measure cognitive ability and job fit.

Website: <https://www.hirevue.com>

Arctic Shores

Markets itself as a task-based hiring platform designed to assess real potential rather than resumes.

Website: <https://www.arcticshores.com>

Harver (formerly Pymetrics)

Uses neuroscience-inspired assessment games to evaluate cognitive and behavioral characteristics.

Website: <https://harver.com>

Examples of Assessments We Can Build

Customer Service Assessment

Scenario:

An angry customer requests a refund while another VIP customer is waiting for support.

Candidate must:

- Prioritize actions
- Decide response approach
- Determine escalation requirements

Measures:

- Empathy
 - Communication
 - Decision Making
 - Customer Judgment
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Sales Assessment

Scenario:

Five leads enter simultaneously.

Candidate must:

- Decide who to contact first
- Allocate limited time
- Maximize revenue opportunity

Measures:

- Commercial Thinking
 - Prioritization
 - Sales Judgment
 - Opportunity Evaluation
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Operations Assessment

Scenario:

Three urgent requests arrive simultaneously.

A deadline is approaching and information is incomplete.

Candidate must:

- Organize tasks
- Make decisions under pressure
- Allocate resources effectively

Measures:

- Problem Solving
 - Logical Thinking
 - Attention to Detail
 - Operational Judgment
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Our Hypothesis

The future of hiring will move away from:

Resume → Interview → Assessment

Towards:

Application → Capability Assessment → Interview

As AI-generated resumes become increasingly common, employers will need new methods to evaluate genuine capability.

Our platform aims to become the infrastructure that enables capability-first hiring by measuring how candidates think, decide, prioritize, and solve problems before employers spend time interviewing them.